Top100 Dataset — General Statistics and Geometric Analysis

Residues: 17,434 | **Chains:** 106 | **Structures:** 98 | **Source:** top100_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

This document provides the following analyses:

1. **General Information** — dataset overview, measurements available, and summary statistics
2. **Amino Acid Composition and Chain Length Distribution** — comparison with UniProt and PDB reference databases
3. **Torsion Angle Distributions** — univariate (backbone, omega deviation, sidechain) and multivariate (Ramachandran) distributions using circular statistics
4. **Bond Angle Distributions** — tau (N−Cα−C) and other non-periodic angles using linear statistics
5. **References**

1. General Information

Dataset	Top100
Total residues	17,434
Unique structures	98
Unique chains	106

Measurements available: phi: 17,218, psi: 17,218, omega: 17,218, tau: 17,434, chi1: 14,380, chi2: 10,616, chi3: 6,441, chi4: 3,099. Not all residues have all measurements — terminal residues lack phi/psi/omega, and glycine and alanine lack sidechain chi angles.

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Ramachandran category distribution:
  General        12,187  (69.90%)
  IleVal          2,052  (11.77%)
  Gly             1,437  ( 8.24%)
  TransPro          720  ( 4.13%)
  PrePro            567  ( 3.25%)
  CisPro             50  ( 0.29%)

Torsion angles (circular statistics):
  Angle             N    Circ Mean     Circ Std
  -------- ---------- ------------ ------------
  phi          17,218      -86.970       43.398
  psi          17,218       27.990      119.824
  omega        17,218      179.521        7.966
  chi1         14,380      -83.355       87.557
  chi2         10,616      174.761      121.133
  chi3          6,441     -171.584      103.209
  chi4          3,099        0.024      102.314

Bond angles (linear statistics):
  tau          17,434      110.729        3.269
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2. Amino Acid Composition and Chain Length Distribution

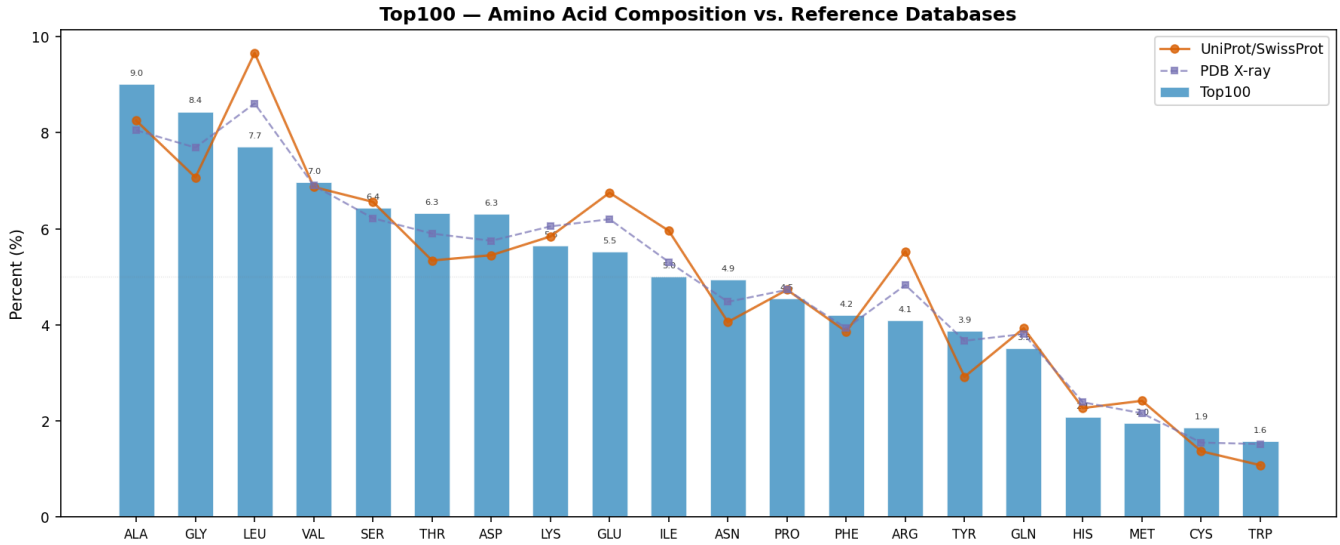
2.1 Representativeness

The figure and table below compare the Top100 amino acid frequencies against two reference databases:

- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

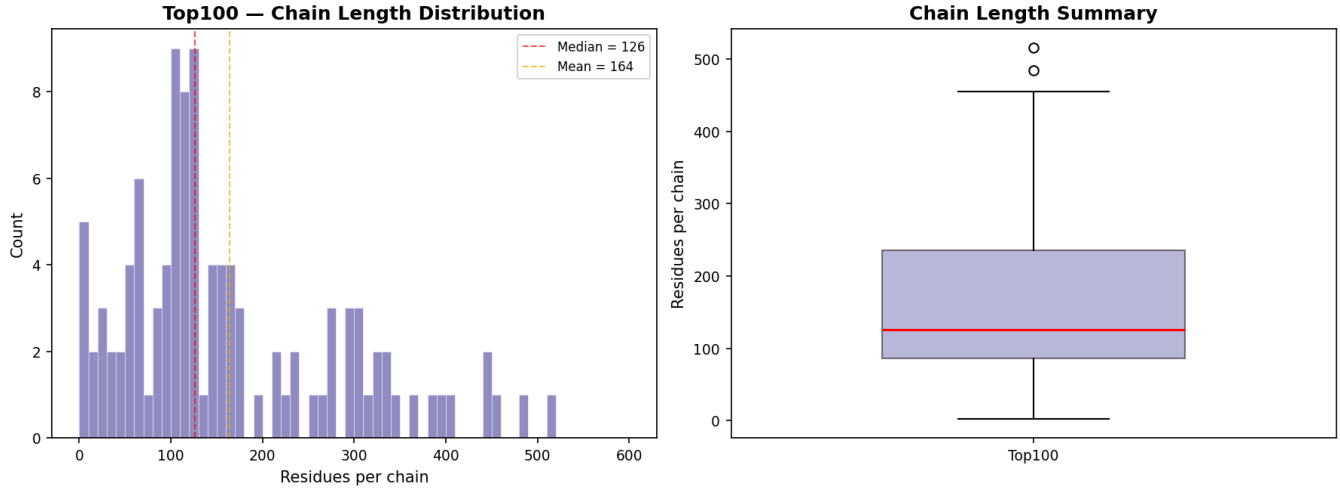
The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition.

The Top100 dataset (17,434 residues from 98 structures) shows additional deviations from both references due to quality filtering, homology reduction, and sample size effects.



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
ALA	1,570	9.01%	8.25%	8.05%	+0.76%
GLY	1,471	8.44%	7.07%	7.69%	+1.37%
LEU	1,343	7.70%	9.66%	8.61%	-1.96%
VAL	1,215	6.97%	6.87%	6.90%	+0.10%
SER	1,122	6.44%	6.56%	6.22%	-0.12%
THR	1,102	6.32%	5.34%	5.90%	+0.98%
ASP	1,099	6.30%	5.45%	5.75%	+0.85%
LYS	985	5.65%	5.84%	6.05%	-0.19%
GLU	963	5.52%	6.75%	6.20%	-1.23%
ILE	872	5.00%	5.96%	5.31%	-0.96%
ASN	862	4.94%	4.06%	4.48%	+0.88%
PRO	792	4.54%	4.73%	4.73%	-0.19%
PHE	732	4.20%	3.86%	3.93%	+0.34%
ARG	714	4.10%	5.53%	4.83%	-1.43%
TYR	676	3.88%	2.92%	3.67%	+0.96%
GLN	613	3.52%	3.93%	3.81%	-0.41%
HIS	364	2.09%	2.27%	2.39%	-0.18%
MET	340	1.95%	2.42%	2.16%	-0.47%
CYS	325	1.86%	1.37%	1.55%	+0.49%
TRP	274	1.57%	1.08%	1.52%	+0.49%

2.2 Chain Length Distribution

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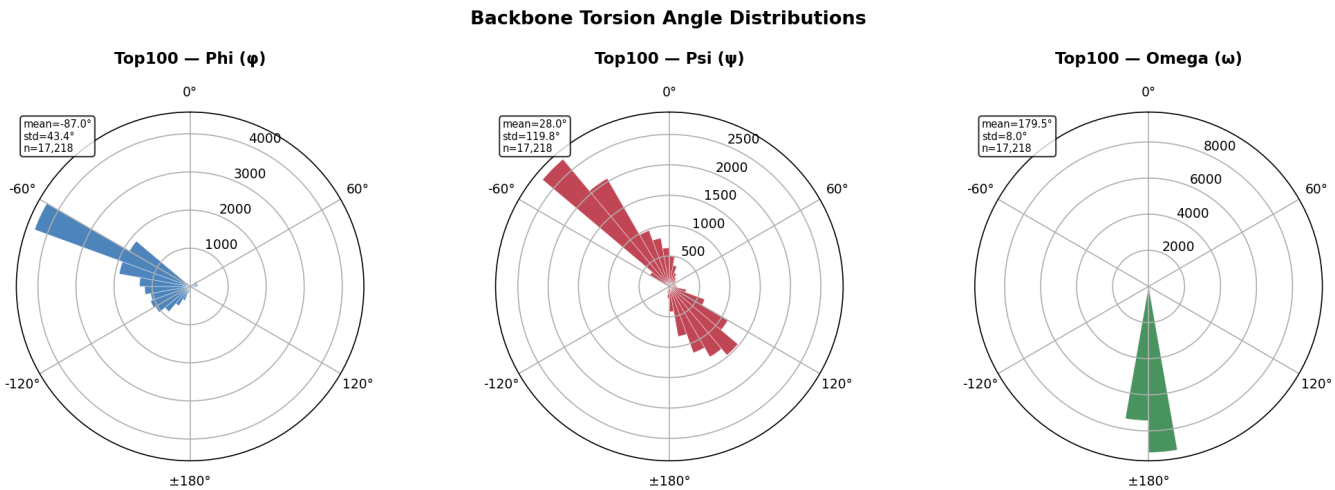
Chain length statistics (106 chains):
  Min:              2
  Q1:              86
  Median:         126
  Mean:         164.5
  Q3:             235
  Max:            516
  Std:          119.8
```

3. Torsion Angle Distributions (Circular Statistics)

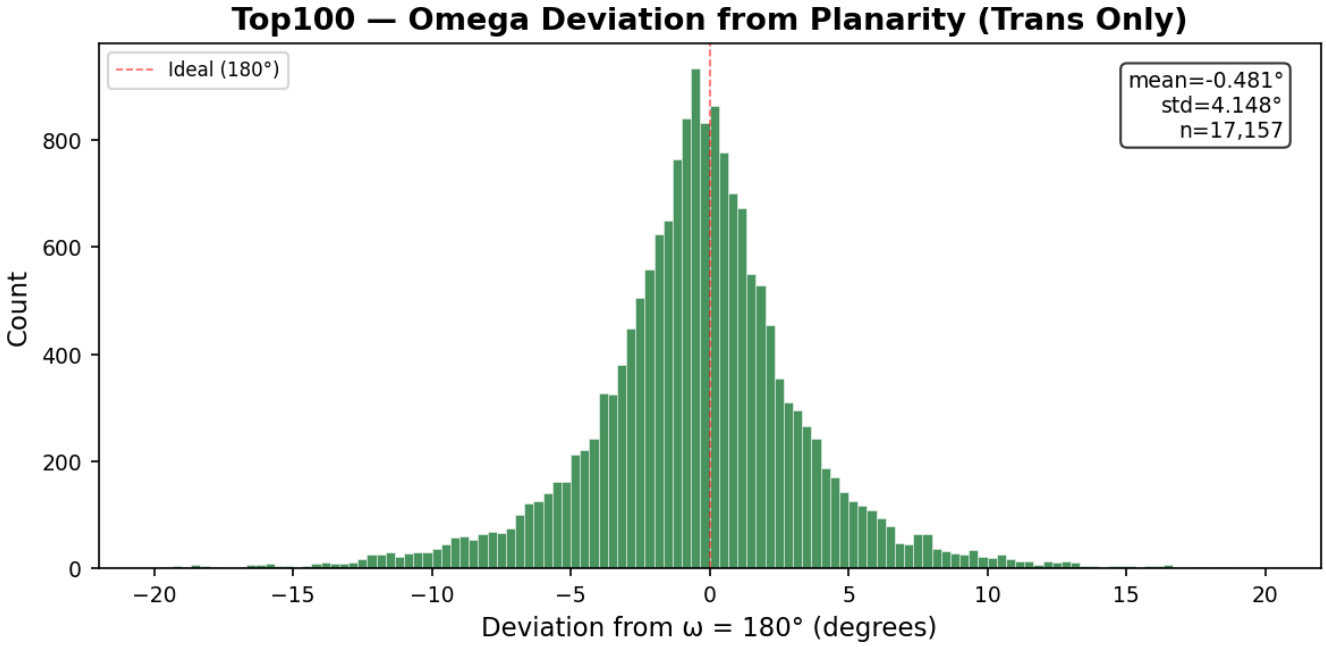
Torsion angles (φ, ψ, ω, χ) are periodic variables defined on the interval [−180°, +180°]. Standard (linear) arithmetic means and standard deviations give misleading results when values cluster near the ±180° boundary — for example, a linear mean of trans peptide bond ω values near +179° and −179° yields ~0° rather than the correct ~180°. All torsion angle summary statistics in this report use **circular statistics**: the circular mean is computed as atan2(⟨sin θ⟩, ⟨cos θ⟩), and the circular standard deviation as $\sqrt{(-2 \ln \bar{R})}$ where $\bar{R}$ is the mean resultant length.

3.1 Univariate Distributions

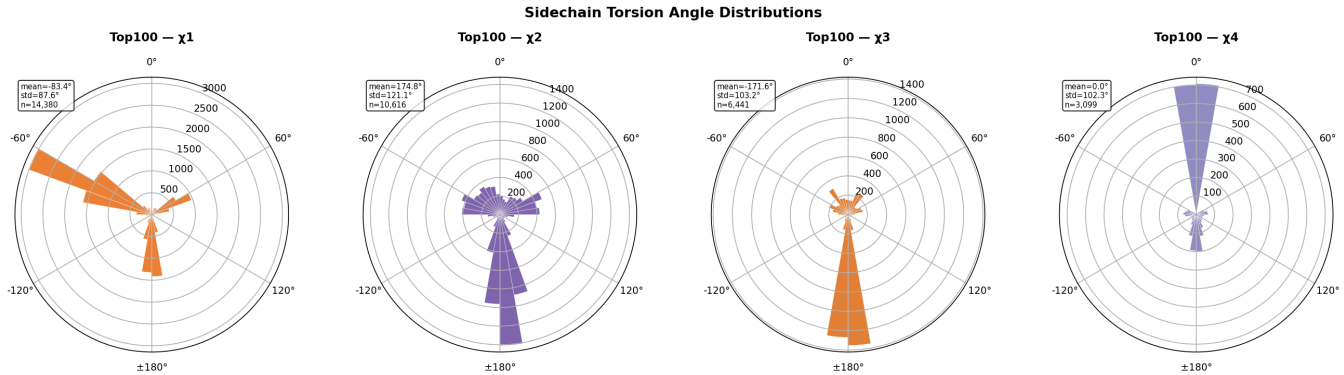
3.1.1 Backbone Torsion Angles (φ, ψ, ω)



3.1.2 Omega Deviation from Planarity (Trans Peptides Only)

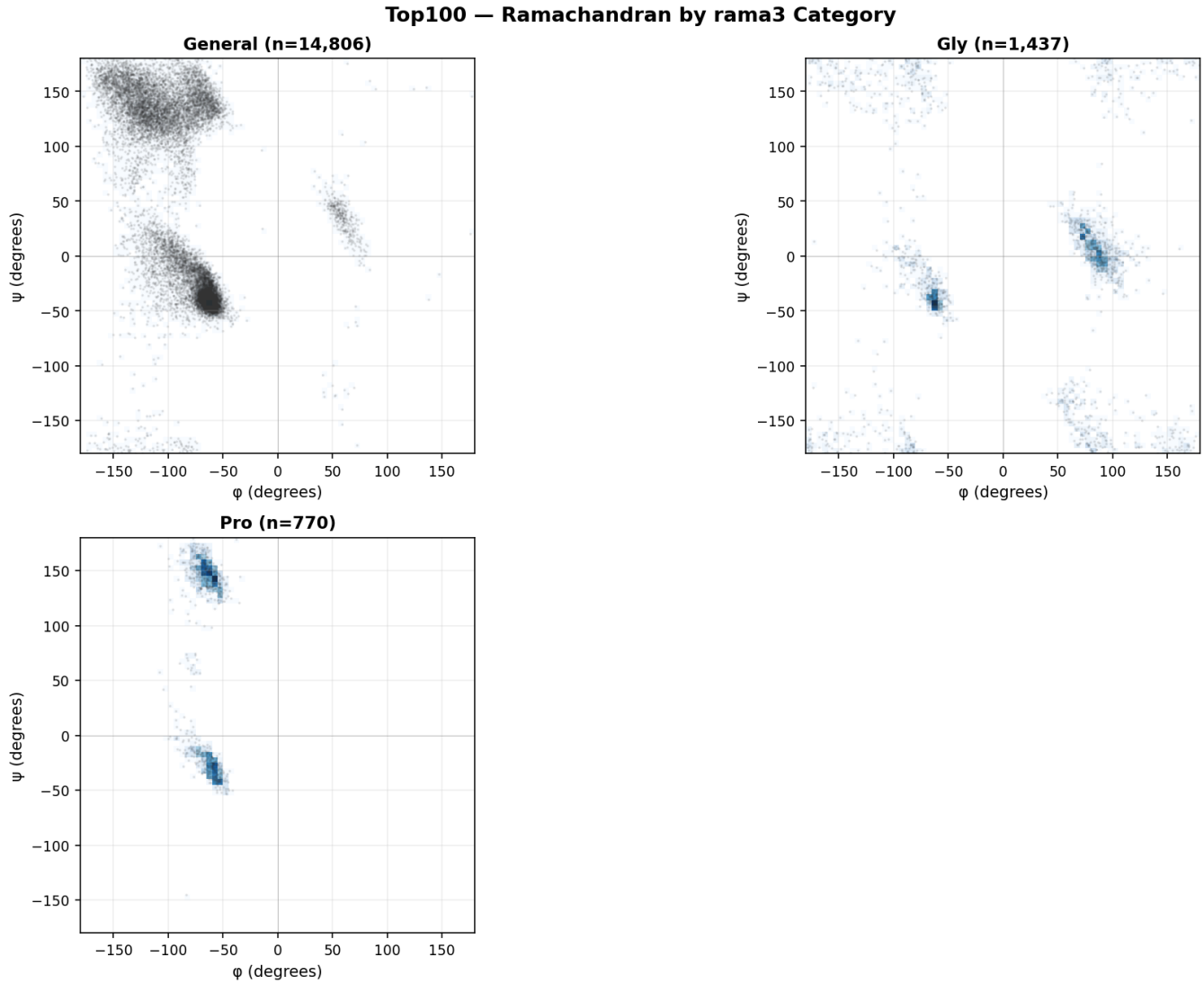


3.1.3 Sidechain Torsion Angles (χ1–χ4)



3.2 Multivariate Distributions

3.2.1 Ramachandran Distributions (φ × ψ by Category)

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Category counts (rama3):
  General        14,806  (84.93%)
  Gly             1,437  ( 8.24%)
  Pro               770  ( 4.42%)
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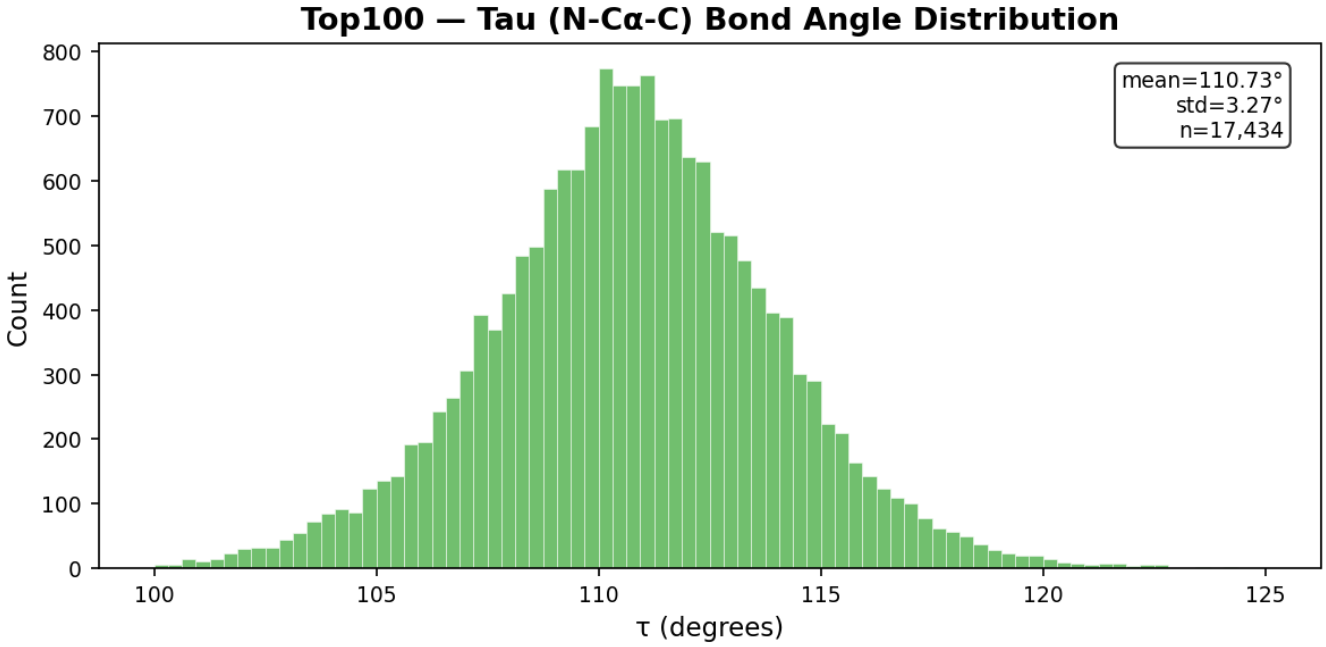
4. Bond Angle Distributions (Linear Statistics)

Bond angles are non-periodic and are analyzed with standard linear statistics (arithmetic mean, standard deviation).

4.1 Tau (N−Cα−C)


Top100 — Tau (N-Cα-C) Bond Angle Distribution
mean=110.73°
std=3.27°
n=17,434
Count
τ (degrees)
0
100
200
300
400
500
600
700
800
100
105
110
115
120
125

5. References

- Lovell, S. C., Davis, I. W., Arendall, W. B. III, de Bakker, P. I. W., Word, J. M., Prisant, M. G., Richardson, J. S., & Richardson, D. C. (2003). Structure validation by Cα geometry: φ, ψ and Cβ deviation. *Proteins*, 50(3), 437–450. doi:10.1002/prot.10286
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